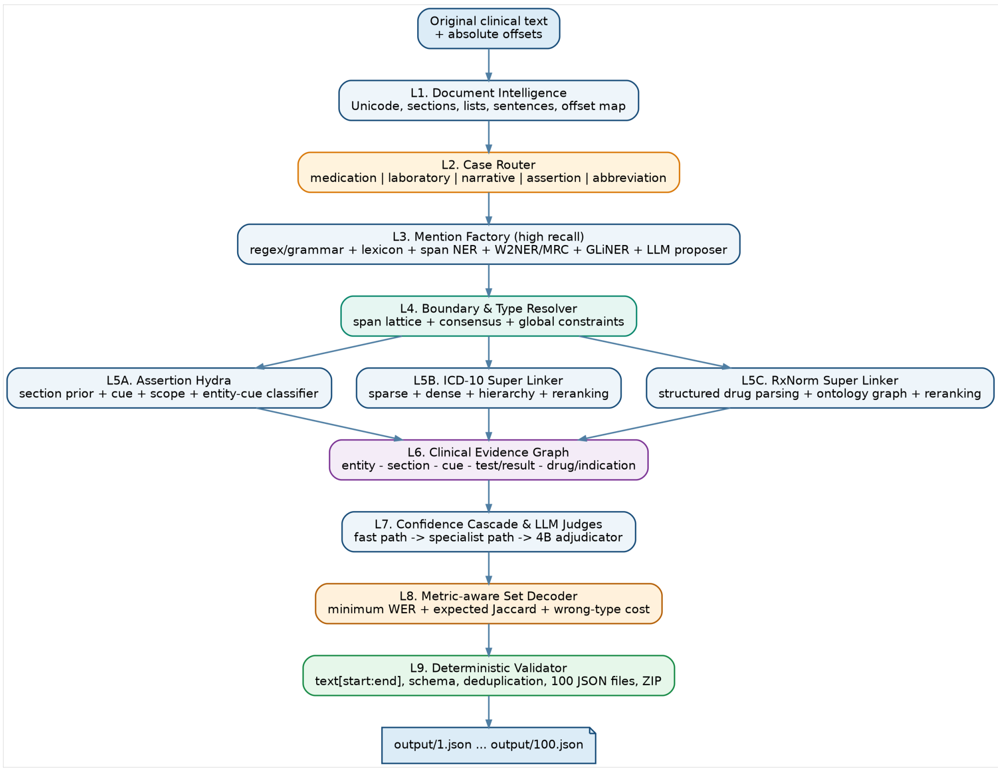
MedNorm-VI

SUPER ARCHITECTURE SPECIFICATION

A Multi-Expert System for Clinical Information Extraction,
Assertion Reasoning, and ICD-10/RxNorm Normalization



Overall MedNorm-VI architecture

Viettel AI Race 2026 - Ontological Reasoning in Medical Knowledge Retrieval

Version 1.1 | 08 July 2026 | Research and implementation design document

Document information and scope

Confirmed constraint: the total number of parameters across all base models in the system must not exceed 9B. Adapter/LoRA parameters are excluded from the budget, so the physical total may be slightly above 9B (for example, approximately 9.1B). This interpretation is used throughout the design.

- **Goal:** design a highly competitive system without artificially limiting implementation complexity or the number of modules.
- **Scope:** the qualification round with 100 input documents and JSON outputs containing `text`, `type`, `assertions`, `candidates`, and `position`.
- **Metric weights:** candidates 40%, text 30%, assertions 30%; a wrong type produces both a false positive and a false negative.
- **Source policy:** prioritize work from ACL, NAACL, EMNLP, TACL, AAAI, NeurIPS, and official WHO/NLM/model-author documentation.
- **Caveat:** this is a research-grounded architecture, not a leaderboard guarantee. Every component must be validated through local evaluation and ablation against the organizer's annotation convention.

Contents

1. Task, metrics, and design consequences

MedNorm-VI converts free-form Vietnamese clinical text into a set of structured entity hypotheses:

```
H = (start, end, text, type, assertions, candidates)
```

Component	Labels/values	Scoring	Architectural consequence
text	Exact substring from the input	1−Word Error Rate	Boundary selection must follow the organizer's convention, not merely preserve meaning.
type	Five entity categories	No direct weight, but wrong type is double-penalized	Use a calibrated classifier with abstention and evidence from grammar and ontology.
assertions	`isNegated`, `isHistorical`, `isFamily`	Jaccard	Avoid extra labels; reason over cue, scope, section, and experiencer.
candidates	ICD-10 codes or RxCUIs	Jaccard, 40% weight	Do not return a broad fixed top-K; calibrate membership and decode expected Jaccard.
position	$[start, end)$	Used for localization/validation	Every normalization must be reversible; an LLM must never calculate offsets freely.

Metric conclusion: candidate mapping has the largest direct weight, but correct entity detection and typing are the gateway. The optimization chain is therefore: entity recall → correct type/boundary → correct assertion/linking → metric-optimal output set.

1.1 Practical objective function

At hypothesis level, MedNorm-VI does not optimize one probability. It estimates multiple confidence families and performs global set selection:

$$S(H) = \alpha S_{span} + \beta S_{type} + \gamma S_{assert} + \delta S_{link} + \varepsilon S_{section} + \zeta S_{consistency} - \lambda Risk_{wrong\_type}$$

After calibration, the decoder selects entities, assertions, and candidate sets that maximize expected score instead of applying a fixed 0.5 threshold.

2. Research synthesis and core principles

P1 - Specialized encoders for spans; LLMs for ambiguity. Clinical NER requires exact token/span boundaries. A multilingual 2024 study found that fine-tuned masked language models can outperform generative prompting in few-shot clinical NER [R06].

P2 - NER should be a multi-representation ensemble. W2NER models word-word relations [R02]; MRC-NER injects type semantics through queries [R03]; T2-NER separates span detection from span-pair classification [R04].

P3 - Linking must be multi-stage and ontology-aware. BioSyn combines sparse/dense retrieval and synonym marginalization [R11]; SapBERT self-aligns synonyms [R12]; KRISSBERT uses knowledge-rich self-supervision and context [R13]; GenBioEL uses KB-guided pretraining and constrained prefix trees [R14].

P4 - Assertion is cue + scope + section + experiencer. NegBERT supports explicit scope modeling [R17], while medSpaCy emphasizes contextual analysis and section detection [R18].

P5 - Pipeline and joint inference should coexist. PURE demonstrates that a strong pipeline can outperform joint entity-relation architectures [R20], while OneIE shows the value of global graph features [R19]. MedNorm-VI uses specialist pipelines followed by a global decoder.

P6 - Synthetic data must be controlled and filtered. Snorkel aggregates noisy labeling functions [R23], and Self-Instruct generates and filters synthetic instruction data [R24]. For NER, generated data must preserve exact spans and the task-specific annotation convention.

P7 - LLMs must not recall ontology IDs from memory. They may only select from retrieved candidates, following constrained generation/KB-guided decoding [R14] and LLM disambiguation after candidate generation [R15].

P8 - Route by case instead of applying one model to every structure. Medication lists, laboratory rows, narratives, assertion scopes, abbreviations, and ambiguous linking expose different structural signals and need different parsers, losses, and confidence rules.

3. MedNorm-VI macro-architecture

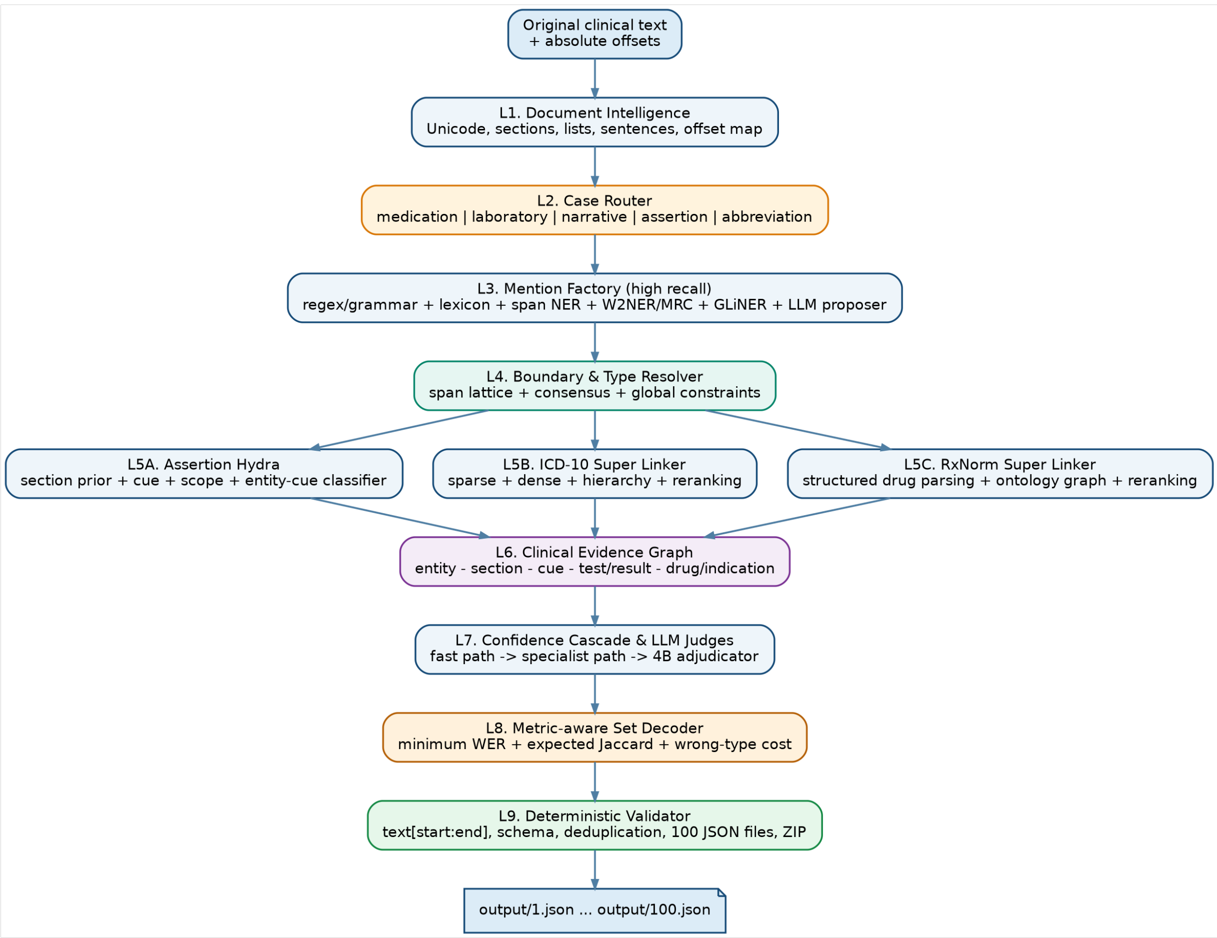


Figure 1. The nine processing layers of MedNorm-VI.

Layer	Responsibility	Primary output	Principle
L1 Document Intelligence	Segmentation, sections, lists, sentences, reversible normalization	DocumentGraph	Never lose original offsets.
L2 Case Router	Assign one or more cases to each segment	Route tags + priors	Multi-label routing; no forced exclusivity.
L3 Mention Factory	Propose every plausible entity span	Span lattice	Optimize recall and preserve provenance.
L4 Boundary & Type Resolver	Select, trim, merge, and type spans	Typed hypotheses	Abstain when wrong-type risk is high.
L5 Specialists	Assertion, ICD, and RxNorm processing	Evidence bundles	Separate logic by ontology and case.
L6 Evidence Graph	Connect entity-section-cue-result-candidate evidence	Clinical graph	Relation reasoning and consistency checks.
L7 Confidence Cascade	Invoke heavy models only when needed	Adjudicated hypotheses	Easy cases bypass the 4B LLM.
L8 Metric Decoder	Select entity/assertion/candidate sets	Optimal prediction set	Expected WER/Jaccard optimization.
L9 Validator	Check schema, offsets, and packaging	100 JSON files + ZIP	Deterministic and fail-fast.

4. Document Intelligence and offset preservation

4.1 Three text representations

Representation	Purpose	May it change?
`original_text`	The only source used to emit `text` and `position`	No.
`normalized_view`	Matching, retrieval, abbreviation expansion, and unit normalization	Yes, but every character must map back.
token/segment graph	Sentences, sections, list items, delimiters, cues, and table-like rows	Yes; every node stores indexes into `original_text`.

```
assert original_text[start:end] == entity["text"]
```

- Perform Unicode normalization on a copy and preserve alignment between original and normalized code points.
- Never trim whitespace or rewrite punctuation before storing a span.
- Split list items while retaining separators and absolute offsets.
- Every span proposal stores `source_model`, `local_score`, route, normalized form, and original coordinates.
- Reject output when text is not an exact substring or the end-index convention is violated.

4.2 Section and discourse parser

A section is evidence, not an absolute rule. For example, "Pre-admission medication list" creates an `isHistorical` prior for medications in that section, but a local sentence such as "started today" may override that prior.

Section pattern	Prior	Benefiting modules
medical history / previous history	`isHistorical`	Assertion, diagnosis typing
pre-admission medications / home medications	`isHistorical`	Medication extraction, RxNorm
family / father / mother / siblings	`isFamily`	Assertion
laboratory / investigations	TEST_NAME + TEST_RESULT	Laboratory parser
diagnosis / conclusion	DIAGNOSIS	Type resolver + ICD
current examination / symptoms	SYMPTOM	Type resolver + assertion

4.3 Expanded lexicons and expression coverage

The patterns above are examples, not closed lists. For history, negation, family context, medication sections, laboratory sections, diagnoses, and symptoms, MedNorm-VI must maintain a **high-coverage expression lexicon**: as many legitimate surface forms as possible, while preserving precision.

- Each meaning is represented by an alias/cue family: full forms, abbreviations, unaccented variants, common misspellings, regional forms, English-Vietnamese code switching, and longer phrase templates. A historical family may include Vietnamese forms equivalent to "history of", "previously", "before admission", "home medications", "diagnosed years ago", and similar expressions.
- Do not rely on exact hard-coded strings alone. Combine exact/regex matching, character n-grams, fuzzy retrieval, abbreviation expansion, and contextual classification to capture unseen formulations.
- Version lexicons by semantic group and store provenance, positive/negative examples, scope, and prior strength. Every change must run regression tests so increased recall does not silently destroy precision.
- Mine new expressions from corpora, validation errors, specialist-LLM disagreements, ontology aliases, and human review before promoting them into the production lexicon.

- A cue only contributes evidence. Scope, section context, and discourse overrides decide which entity actually receives an assertion; keyword presence must never label every entity indiscriminately.

Group	Coverage examples	Additional mechanism
Historical	history of, previously, before admission, home medication, diagnosed since, chronic for years	Lexicon + section patterns + temporal classifier
Negation	no, not yet, denies, no evidence of, ruled out, negative for	Cue lexicon + scope/syntax model
Family	family, father, mother, sibling, relative, maternal/paternal side	Experiencer lexicon + dependency/discourse relation
Medication section	home meds, background therapy, pre-hospital meds, old prescription, current medication list	Header variants + layout/list features
Laboratory	laboratory, investigations, hematology, chemistry, ancillary results	Section aliases + table-like structure

5. Case Router and specialized flows

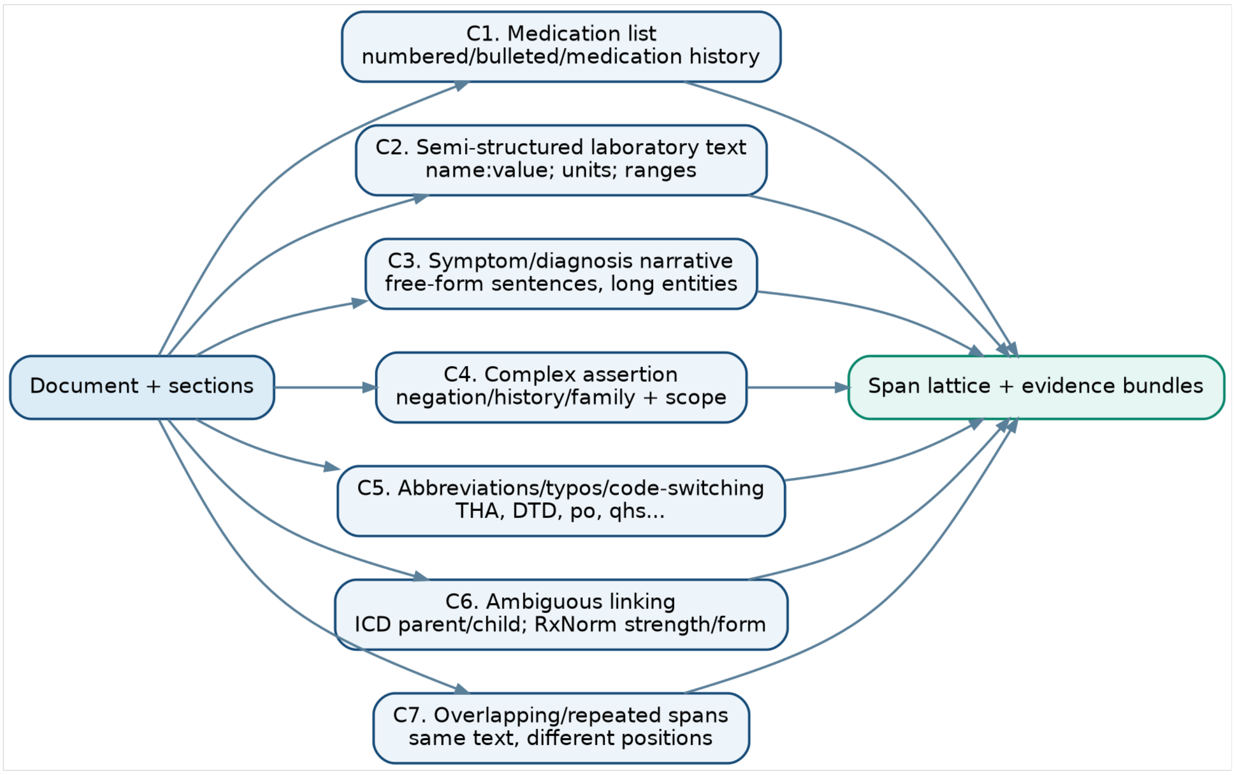


Figure 2. A multi-label router sends each segment to the appropriate specialists.

The router does not classify an entire document into exactly one type. A segment may simultaneously carry C1 + C4 + C6, for example a pre-admission medication list containing a drug with ambiguous strength.

Case	Signals	Preferred flow
C1 Medication-list	Numbering/bullets, repeated route-frequency patterns, repeated drug names	Medication grammar + section propagation + structured RxNorm linker
C2 Lab semi-structured	Colons/semicolons, units, decimals, reference ranges	Finite-state parser + test-result bipartite matching
C3 Clinical narrative	Long sentences, multiple symptoms/diagnoses, soft boundaries	Span-NER ensemble + MRC/W2NER + contextual type scorer
C4 Assertion-heavy	Negation/history/family cues, conjunctions and contrast	Cue detector + scope model + section prior + relation classifier
C5 Abbreviation/noise	THA, DTD, qhs, accent errors, code switching	Abbreviation lattice + character retrieval + LLM expansion verifier
C6 Ambiguous linking	ICD parent/child; same RxNorm ingredient with different strength/form	Ontology hard negatives + cross-encoder + constrained judge
C7 Overlap/duplicates	Same text at multiple positions, nested/adjacent spans	Absolute offsets + interval/graph optimizer + no text-only deduplication

6. Mention Factory: high-recall span generation

The Mention Factory produces a unified span lattice from multiple specialists. No individual module is allowed to emit a final entity directly.

Expert	Method	Strength	Expected failure
E1 Medication grammar	Lexicon + finite-state grammar for ingredient/salt/release/strength/form/route/frequency	Long deterministic medication boundaries	Novel names, severe typos
E2 Lab parser	Regex/grammar for name:value-unit-range	High precision on laboratory records	Narrative laboratory text
E3 ViHealthBERT span model	Span enumeration + biaffine/boundary head	Vietnamese clinical language	Novel entities
E4 PhoBERT W2NER	Word-word grid relations [R02]	Overlap and discontinuous-like patterns	$O(n^2)$ cost and data demand
E5 XLM-R MRC-NER	Type-specific queries [R03]	Type semantics and code switching	Slower than token classification
E6 GLiNER medium/large	Open-type proposals [R28]	Unseen surface forms	Not trained for the organizer's annotation style
E7 Qwen3-1.7B proposer	Substring/anchor proposals for hard cases	Long context and abbreviations	Hallucination; proposal-only role

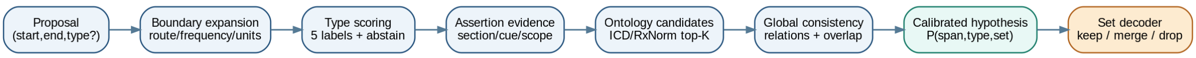


Figure 3. Lifecycle of a hypothesis from proposal to set-level decision.

6.1 Detailed medication grammar

```
MED = NAME [SALT] [RELEASE] [STRENGTH|CONCENTRATION] [DOSE_FORM] [ROUTE] [FREQUENCY] [PRN]
```

- The name lexicon includes ingredients, brands, salts, and unaccented variants.
- The strength parser supports ranges (325–650 mg), ratios (0.4 MG/ML), decimal commas, and multi-ingredient products.
- Release forms: IR/ER/XR/XL/SR/CR; routes: po/iv/im/sc/oral; frequencies: daily/bid/tid/qid/qNh/qam/qhs/prn.
- Multiple boundary candidates are generated; the resolver learns the organizer's convention instead of allowing the grammar to decide unilaterally.

6.2 Laboratory parser and relation proposals

```
TEST_NAME -has_result-> VALUE [UNIT] [FLAG] [REFERENCE_RANGE]
```

Test-result pairing is a matching problem whose cost incorporates distance, delimiter type, shared row/list item, and unit compatibility. This internal relation helps typing and boundary resolution even though round-one output does not contain a relation field.

7. Boundary & Type Resolver

The resolver receives overlapping spans and builds a feature vector for each hypothesis:

$$\phi(H) = [\text{model logits, grammar completeness, lexicon match, section prior, boundaries, ontology evidence, route features}]$$

7.1 Boundary ensemble

- A boundary-offset head predicts left/right expansion or contraction, inspired by boundary-aware and offset-prediction NER [R05].
- Hard negatives omit strength/route/frequency or swallow delimiters/list numbers.
- The loss combines exact-span loss, token IoU, and a WER surrogate to distinguish near-miss spans.
- Medication-list routes weight grammar completeness more heavily; narrative routes weight contextual encoders more heavily.

7.2 Type scoring and abstention

Evidence	Example	Effect on type
Morphology/grammar	mg, ml, po, q6h	Increase MEDICATION
Section	Diagnosis: ...	Increase DIAGNOSIS
Ontology hit	Exact RxNorm strength match	Increase MEDICATION
Local predicate	diagnosed with / has	Increase DIAGNOSIS
Treatment purpose	treated for cough	Increase SYMPTOM for "cough"
Colon + numeric	WBC:14.43	TEST_NAME / TEST_RESULT

Abstention: because a wrong type is double-penalized, a hypothesis with high entropy or a strong type-ontology conflict may be dropped instead of guessed. Thresholds are optimized on the local evaluator.

7.3 Global span optimization

A weighted interval/graph optimizer selects the highest-utility span set subject to hard and soft constraints:

- Never deduplicate by text alone; merge only when position and semantic identity are the same.
- Permit nested spans only if the annotation/data support them.
- Penalize near-complete overlap between two spans of the same type.
- Preserve TEST_NAME–RESULT pairs with a strong `has_result` edge.
- Forbid ICD candidates on MEDICATION and RxNorm candidates on DIAGNOSIS.

8. Assertion Hydra

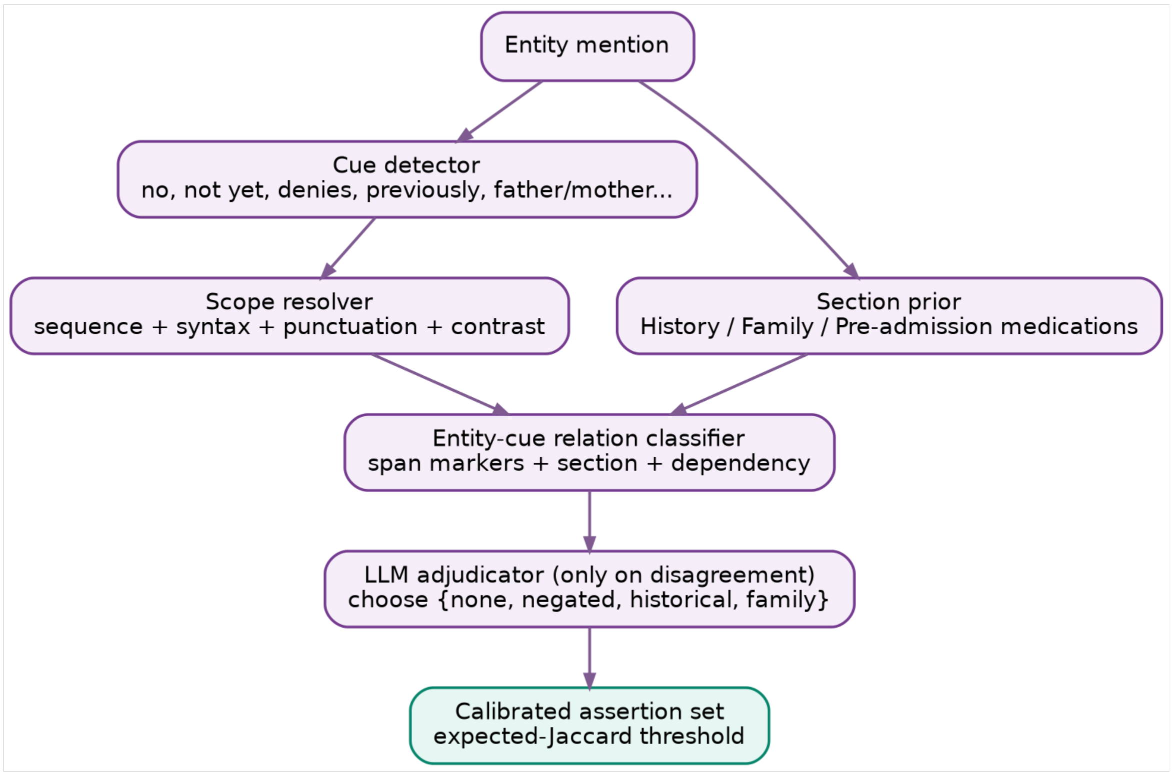


Figure 4. Assertions are resolved using section, cue, scope, relation, and adjudication.

Stage	Model/logic	Output
A1 Section prior	Rules + learned section classifier	Section-level P(historical/family)
A2 Cue detector	ViHealthBERT token classifier + broad lexicon	Cue spans and cue types
A3 Scope resolver	Sequence labeling + syntax/punctuation features	Token scope for each cue
A4 Entity-cue classifier	Span markers, distance, section, scope overlap	P(assertion | entity,cue)
A5 LLM adjudicator	Qwen3 constrained multiple choice	Used only on rule/model disagreement
A6 Set calibration	Per-label temperature/isotonic + expected Jaccard	Final assertion set

8.1 Vietnamese clinical scope rules

- A Vietnamese negation cue equivalent to “not” usually stops at contrast markers such as “but” or “however”, unless the learned scope model provides contrary evidence.
- Coordinated forms equivalent to “not A, not B” create two scopes; “not A or B” requires syntax/context classification.
- A history cue may scope over an entire noun phrase, list, or section.
- A family experiencer may occur far from the entity in the subject; use dependency and discourse edges.
- An entity may carry multiple assertions; decoding is multi-label, not exclusive softmax.

9. ICD-10 Super Linker

The ICD linker handles DIAGNOSIS entities. The knowledge base must be frozen to the exact organizer-provided version; WHO ICD-10 provides hierarchy and official downloads [R34,R35].

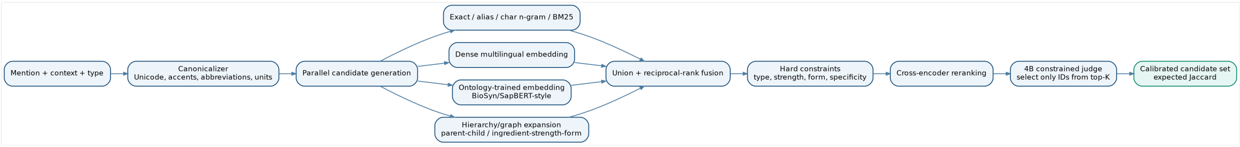


Figure 5. Linking architecture with multiple retrievers, constraints, and a constrained judge.

9.1 Multi-representation indexes

Index	Content	Goal
Exact alias	Canonical labels, synonyms, abbreviations, Vietnamese variants	High precision
Character n-gram	Typo and unaccented surface forms	Robustness
BM25/sparse	Labels, definitions, inclusion terms	Lexical recall
BGE-M3	Dense+sparse+multi-vector multilingual retrieval [R16]	Semantic recall
Qwen3 Embedding 0.6B	Instruction-aware multilingual embedding [R31]	Cross-lingual/complex phrases
SapBERT/BioSyn-style	Ontology-synonym contrastive space [R11,R12]	Fine-grained medical synonymy
Hierarchy graph	Parent, child, sibling, chapter	Specificity reasoning

9.2 Candidate generation and reranking

1. Canonicalize the mention and generate query variants: accented/unaccented, abbreviation expansions, and Vietnamese-English medical aliases.
2. Query indexes in parallel and merge with route-aware weighted reciprocal-rank fusion.
3. Expand top candidates through parent/child/sibling relations while retaining graph provenance.
4. Cross-encoder rerank on mention + sentence + section + candidate label/definition/hierarchy.
5. Qwen3-4B receives only the top 5–10 and may select only valid IDs; it cannot invent a code.
6. Candidate-set decoding returns one or multiple codes using calibrated membership probabilities and expected Jaccard.

9.3 Specificity controller

The controller chooses a general or specific code based on textual evidence. A child code is penalized when it requires an absent attribute; a parent is penalized when the text already provides enough specificity. This hierarchy-aware decision cannot be reduced to cosine similarity alone.

10. RxNorm Super Linker

RxNorm provides normalized drug names and RxCUIs. NLM distinguishes term types such as SCD (ingredient + strength + dose form) and SBD (plus brand) [R32,R33], so structured parsing precedes retrieval.

10.1 Structured drug representation

```
D = {ingredient[], salt, strength[], unit, concentration, dose_form, release, brand, route,
frequency, prn}
```

Field	Role in linking	Constraint
ingredient	Primary identity	Hard when confidence is high
strength + unit	Distinguish RxCUIs with the same ingredient	Hard / very high penalty
dose form	Tablet, solution, suspension, injection, etc.	High penalty
release	IR/ER/XR/XL/SR	High penalty
brand	Select SBD instead of SCD	Conditional
route/frequency/PRN	Usually not RxCUI identity, but useful for boundary/context	Never a hard identity feature

10.2 Graph search in RxNorm

Candidate generation is not name search alone. The system traverses relations among ingredient, clinical drug component, semantic clinical drug, brand, and dose form. If an exact SCD is unavailable, the graph path yields the closest valid candidate and explains which field is missing.

- Exact structured match has highest priority.
- Same ingredient but different strength is the most important hard negative.
- Same strength but different release/dose form is the second-level hard negative.
- Brand/generic handling follows TTY and the organizer's annotation convention.
- Historical or suppressed concepts are filtered according to the frozen KB snapshot if only active concepts are accepted.

11. Clinical Evidence Graph and global reasoning

All evidence is merged into an internal graph. Nodes include spans, sections, cues, list items, laboratory values, and ontology candidates. Edges include `has_result`, `modified_by`, `in_section`, `treats`, `overlaps`, `same_surface`, and `candidate_of`.

Constraint/feature	Example	Effect
Section consistency	Medication inside “pre-admission”	Historical prior
Test-result pairing	WBC ↔ 14.43	Preserve correct type and boundary
Drug-indication	guaifenesin ... for cough	“cough” is a symptom, not a drug
Candidate-type consistency	Exact RxCUI match	Reinforce MEDICATION type
Overlap competition	amlodipine vs. amlodipine 10 mg po daily	Select by expected WER
Repeated mention	constipation appears twice	Keep two nodes with different offsets

Global inference may use beam search with global features in the spirit of OneIE [R19], or ILP/weighted graph optimization. The first implementation should favor beam search/weighted intervals for debuggability; ILP becomes a research branch as constraints grow.

12. Confidence Cascade and LLM committee

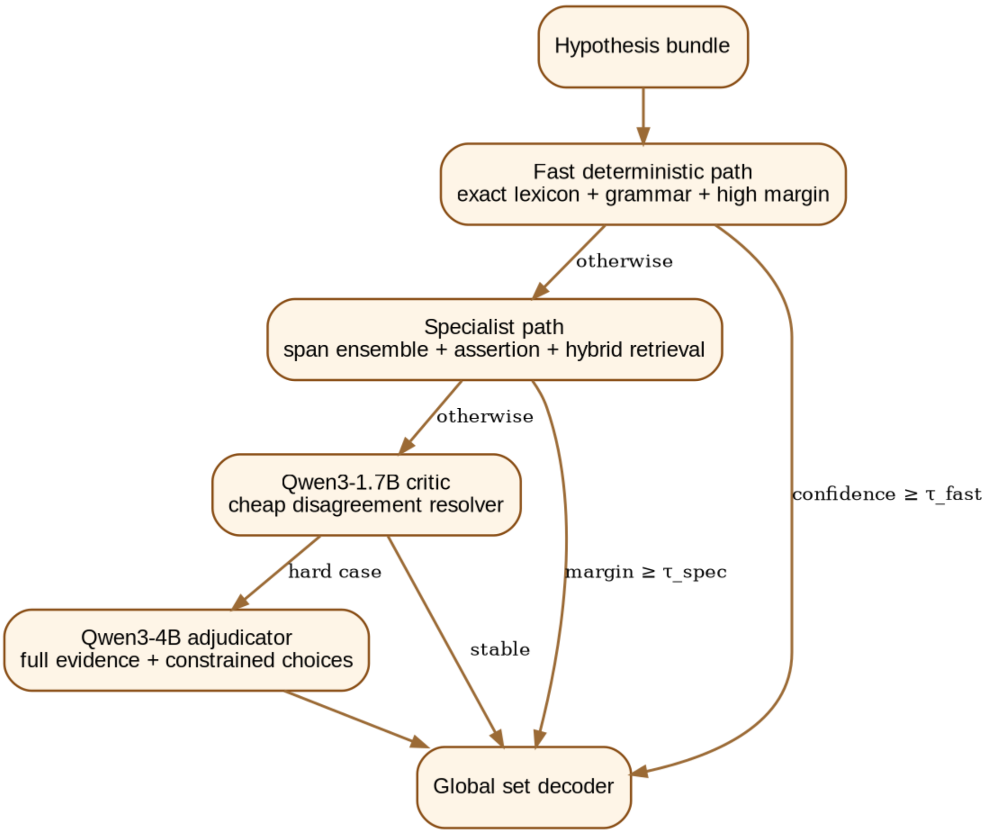


Figure 6. Only difficult hypotheses reach Qwen3-4B.

The cascade reduces hallucination and confines LLMs to adjudication. Each stage returns a structured decision and confidence, never the final document JSON.

Level	Entry condition	Model	Task
Fast path	Exact grammar/alias + high margin	Rules + specialist encoders	Accept directly
Specialist path	Span/type/linker disagreement	NER ensemble + hybrid retrieval	Build evidence bundle
Critic	Two close alternatives	Qwen3-1.7B	Critique boundary/type/assertion/top candidates
Adjudicator	Hard case / high metric impact	Qwen3-4B	Select from a locked option set
Global decoder	All hypotheses calibrated	No LLM required	Select final set

12.1 Prompt contract

```
INPUT:
- original sentence + section
- candidate span(s) with offsets
- specialist scores and triggered rules
- top ontology candidates with canonical names/attributes

OUTPUT (constrained):
{"choice":"A|B|C|NONE", "assertions":[...], "confidence":0..1}
```

- The LLM cannot change start/end unless that span is present in the option set.
- It cannot emit ICD/RxCUI values outside retrieved top-K.
- Do not store chain-of-thought; store only the final structured decision and auditable evidence.
- Qwen3-4B supports thinking/non-thinking modes and has approximately 4.0B parameters [R29]; Qwen3-1.7B is the cheaper critic [R30].

13. Metric-aware Set Decoder

13.1 Text and boundary

For competing spans, the decoder estimates expected utility under WER. It considers length, missing tokens, and annotation priors rather than selecting the highest raw probability.

$$U_{text}(span) \approx E[1 - WER(span, gold\_span) \mid evidence]$$

13.2 Candidate set

If each candidate c has probability $p(c \in G)$, search a small family of prefixes/top combinations for the set that maximizes expected Jaccard. Top-1 is not always optimal, and fixed top-K is not used.

$$P^* = \arg\max_P E[|P \cap G|/|P \cup G|]$$

13.3 Assertion set

Assertions are multi-label. Because an empty ground truth with an extra predicted label receives zero Jaccard, each label threshold must be calibrated separately under class imbalance and cost.

13.4 Entity existence/type

The utility of retaining an entity subtracts wrong-type risk. A near-correct span with uncertain type may be more harmful than omission; thresholds are learned using the local evaluator through search or Bayesian optimization.

14. Data Engine and self-improvement loop

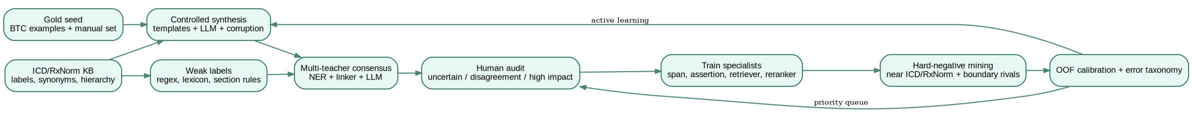


Figure 7. Data flywheel combining ontologies, weak supervision, teachers, and human audit.

14.1 Data sources

Source	Labels produced	Control
Organizer examples + manual gold	Full schema	Primary annotation convention source
ICD/RxNorm ontologies	Aliases, candidate pairs, hierarchy	Exact organizer snapshot and license
Public Vietnamese health corpora	Language/domain adaptation	No data directly related to private test
Synthetic templates	Span/type/assertion/linking	Exact offsets generated by construction
Weak labeling functions	High-volume pseudo-labels	Snorkel-style label model [R23]
Teacher consensus	Cases outside templates	Retain only high-agreement/high-evidence samples

14.2 Controlled synthesis by case

Case generator	Required variation
Medication list	Numbering, brand/generic, strength ranges, route/frequency, historical headers, typos
Laboratory	Decimal comma/dot, units, missing units, H/L flags, reordered columns, multiline formats
Narrative	Coordination, apposition, symptom/diagnosis ambiguity, long spans
Assertion	Negation scope, contrast, family experiencer, historical sections, multiple labels
Linking hard negatives	ICD parent-child/sibling; RxNorm same ingredient with different strength/form/release
Boundary corruption	Drop/add left/right tokens, delimiters, list numbers, treatment phrases

14.3 Active-learning priority

```
priority = disagreement * metric_weight * uncertainty * novelty * frequency
```

Prioritize cases with high candidate-weight impact, high wrong-type risk, or disagreement among specialists and LLMs. Do not select samples by entropy alone.

15. Training strategy

Stage	Model	Objective	Data
S0 Domain adaptation	ViHealthBERT/PhoBERT/XLM-R	MLM / continued pretraining	Vietnamese health text + ontology descriptions
S1 Mention extraction	Span/W2NER/MRC heads	Span BCE/focal + boundary offset + type loss	Gold + synthetic + weak
S2 Assertion	Cue/scope/relation heads	Multi-task token + pair classification	Controlled assertion corpus
S3 Retrieval	BGE/Qwen embeddings/SapBERT-style	Contrastive + synonym alignment + hard negatives	ICD/RxNorm aliases
S4 Reranking	Qwen reranker/cross-encoder	Pairwise/listwise ranking	Retrieved hard negatives
S5 LLM critic/judge	Qwen3 LoRA	Constrained classification/SFT	Evidence bundles + teacher/manual decisions
S6 Calibration/stacking	Small meta-model	NLL/Brier + score optimization	Out-of-fold predictions

15.1 Multi-task sharing

To save budget and increase transfer, ViHealthBERT and XLM-R backbones are loaded once with multiple adapters/heads for span, boundary, cue, scope, and type. The organizer excludes adapters from the 9B budget according to the team's confirmation; every adapter must still appear in the manifest.

15.2 Cross-validation against leakage

- Group-split by template/source/document so synthetic variants cannot cross folds.
- Manage ontology aliases of the same concept when measuring zero-shot concepts.
- Calibration and threshold search use out-of-fold predictions only.
- Every leaderboard decision records config, seed, Git commit, and artifact hash.

16. End-to-end inference flow

1. Read UTF-8 input while preserving the exact decoded text required by the task.
2. Build the DocumentGraph: sections, sentences, list rows, table-like cells, and absolute offsets.
3. Assign route tags and priors to each segment with the Case Router.
4. Run grammar/lexicon parsers and NER specialists to create the span lattice.
5. Resolve boundaries/types and remove hypotheses with low expected utility.
6. Generate assertion evidence and calibrated ICD/RxNorm candidate distributions.
7. Add relations and consistency features through the Clinical Evidence Graph.
8. Adjudicate only uncertain or high-impact cases through the Confidence Cascade.
9. Select the metric-optimal entity/assertion/candidate set.
10. Validate substrings, schema, allowed labels/codes, duplicates, ordering, and create the ZIP.

Fail-fast: if any entity violates `original_text[start:end] == text`, or a code is absent from the frozen KB snapshot, stop that file and write an audit log. Never silently repair output.

17. Total 9B parameter budget

The "Safe Submission Stack" uses several base models while keeping the total below 9B. Heads/adapters are outside the budget according to organizer confirmation. Counts are approximate and must be verified by an automated parameter-count script before submission.

Base model	Approx. params	Role
Qwen3-4B	4.000B	Hard-case adjudicator
Qwen3-1.7B	1.700B	Fast critic/proposer
Qwen3-Embedding-0.6B	0.600B	Multilingual dense retrieval
Qwen3-Reranker-0.6B	0.600B	Cross-encoder/listwise reranking
BGE-M3	0.568B	Dense+sparse+multi-vector retrieval
XLM-R large	0.560B	MRC/type/code-switch backbone
GLiNER medium v2.1	0.209B	Open-type mention proposals
PhoBERT-large	approximately 0.370B	Vietnamese W2NER/boundary model
ViHealthBERT	approximately 0.135B	Vietnamese clinical span/assertion backbone
SapBERT/BiomedBERT base	approximately 0.110B	Ontology-synonym embedding
TOTAL BASE	**approximately 8.852B**	Below 9B; adapters/heads excluded by organizer confirmation

Compliance recommendation: operate near 8.85B to retain metadata safety margin. GLiNER-large 459M is a research alternative; if used, remove the separate SapBERT/BiomedBERT encoder so the base total remains approximately 8.99B. Adapter/LoRA parameters may make the physical total slightly exceed 9B but are outside the confirmed base-model budget.

17.1 Proposed submission configurations

Variant	Change	Expected total	Use
Safe-8.85B	GLiNER medium 209M instead of large 459M	approximately 8.85B	Default submission
Alt-8.99B	GLiNER large 459M; remove separate SapBERT/BiomedBERT	approximately 8.99B	Higher open-type recall, weaker independent ontology encoder
Lean-6.5B	Remove Qwen3-1.7B, GLiNER, and separate SapBERT	approximately 6.4–6.6B	Debugging/Colab/ablation
No-LLM baseline	Remove both generative Qwen models	approximately 3.4B	Measure the real contribution of LLMs

18. Evaluation, ablation, and error analysis

18.1 Mandatory local evaluator

- Reproduce text score, assertion Jaccard, candidate Jaccard, and final weighted score.
- Report by type, section, route, entity length, abbreviation status, and seen/unseen concept.
- Separate missed entity, spurious entity, wrong type, left/right boundary, cue/scope, retrieval miss, reranking miss, and set-decoding error.
- Export HTML/JSON audits for each sample with complete provenance and scores.

18.2 Ablation matrix

Ablation	Question
Grammar only → +NER	Which cases gain recall, and where does precision fall?
Single NER → ensemble	Do W2NER/MRC/GLiNER contribute complementary evidence?
No section prior	How much assertion score comes from section propagation?
Sparse → dense → ontology → reranker	What is the candidate gain at each stage?
No LLM / critic / adjudicator	Do LLMs improve hard cases or damage deterministic ones?
Top-1 vs fixed top-K vs expected-Jaccard	Does set decoding improve candidate score?
Local selection vs global graph	How much do constraints reduce duplicates and wrong types?
Synthetic volume/quality	When does synthetic data create domain or annotation drift?

18.3 Stress suites

- Offset stress: decomposed Unicode, repeated spaces, newlines, punctuation, duplicate substrings.
- Assertion stress: coordination, contrast, nested history/family context, double negation.
- RxNorm stress: same ingredient/different strength, concentrations, release, brand/generic.
- ICD stress: parent/child, acute/chronic, with/without complication, vague diagnosis.
- Noise stress: unaccented Vietnamese, typos, mixed English abbreviations, OCR-like artifacts.
- Packaging stress: missing/extra files, malformed JSON, wrong names/order/encoding.

19. Repository architecture and module contracts

```
MedNorm-VI/
|-- configs/              # configs and parameter-budget manifest
|-- data/                 # raw, KB snapshots, synthetic, gold, folds
|-- mednorm/
|   |-- document/         # normalization, offsets, sections, list/table parser
|   |-- routing/          # case router
|   |-- mentions/         # grammar, lexicon, span/W2NER/MRC/GLiNER
|   |-- resolution/       # boundary, type, overlap/global optimizer
|   |-- assertions/       # cue, scope, section, relation, calibration
|   |-- linking/icd/      # indexes, graph, retrieval, reranking
|   |-- linking/rxnorm/   # structured parser, graph, retrieval, reranking
|   |-- graph/            # evidence graph and global features
|   |-- llm/              # critic/adjudicator, constrained schemas
|   |-- decoding/         # expected WER/Jaccard set decoder
|   |-- validation/       # schema, offset, KB, packaging
|   `-- audit/            # provenance, reports, replay
|-- scripts/              # prepare/train/index/infer/evaluate/package
|-- tests/                # unit, golden, stress, reproducibility
|-- weights/              # local/self-hosted checkpoints + adapters
|-- outputs/
|-- Dockerfile
|-- requirements.lock
`-- README.md
```

19.1 Core data contracts

`SpanProposal:`
start, end, text, proposed_types, source, local_score, route, features

`EntityHypothesis:`
span, type_distribution, assertions_distribution, icd_distribution, rxnorm_distribution, evidence_ids, calibrated_score

`FinalEntity:`
text, type, assertions, candidates, position

- Modules should be pure-ish functions over an immutable DocumentGraph whenever possible.
- Every decision stores provenance for replay and ablation.
- Linking modules may not silently edit spans; they may only emit scored boundary feedback.
- Configs store snapshot IDs, model hashes, thresholds, and parameter counts.

20. Implementation roadmap

Phase	Deliverables	Exit criterion
Phase 0 - Contract & evaluator	Schema, offsets, metric clone, stress tests, audit format	Complete before writing models
Phase 1 - Deterministic baseline	Document parser, medication/lab grammar, lexicon, exact/fuzzy linking	Valid `output.zip`
Phase 2 - Specialist encoders	ViHealthBERT/PhoBERT span+type+assertion	Out-of-fold predictions
Phase 3 - Super linkers	ICD/RxNorm multi-index, graph, hard negatives, reranker	Recall@K and Jaccard calibration
Phase 4 - Global resolver	Span lattice, constraints, evidence graph, metric decoder	Fewer wrong types/duplicates
Phase 5 - LLM cascade	1.7B critic, 4B constrained adjudicator, LoRA	Hard cases only
Phase 6 - Data flywheel	Synthetic generators, weak labels, active learning	Broader annotation coverage
Phase 7 - Reproducibility	Docker, offline weights, one-command inference, source package	Ready for private-test rebuild

Order must not be reversed: evaluator, offset logic, and validator must be completed before the heavy architecture. Otherwise model changes cannot be judged against the organizer's true metric.

21. Risks, controls, and unresolved decisions

Risk	Impact	Control
Insufficient annotation examples	Medically correct but organizer-incompatible spans/types	Small manual gold set + rulebook + error-driven updates
Different ICD/RxNorm snapshots	Correct code in another version but scored wrong	Use organizer KB and freeze hashes
Synthetic drift	Artificial validation gain, hidden-test degradation	Source-group split, human audit, realistic stress tests
LLM hallucination	Span/code outside input/KB	Constrained options + validator + no free generation
Total parameters exceed 9B	Invalid submission	Automated parameter manifest + safety margin

Risk	Impact	Control
Colab VRAM/time	Full stack cannot reside in memory simultaneously	Sequential load/unload, offline caching, batch by module
Wrong-type overprediction	Double penalty	Abstention, calibration, type-link consistency
Low retrieval recall	Reranker cannot recover missing candidates	Multi-index + query expansion + alias/hard-negative mining
Private rebuild failure	Disqualification	Docker, lockfile, local weights, one command, audit logs

21.1 Decisions requiring organizer data

- Does entity matching use position, or only text + type?
- What exact tokenization/normalization is used for WER?
- Which ICD-10 version and RxNorm release are provided?
- Is the candidate list length limited, and does ordering matter?
- Are nested/overlapping entities present?
- Will later rounds include an explicit relation field?
- Are there time/memory limits for private rebuild?

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Appendix A - Checklist before the private test

☐ Parameter-count script confirms base total and adapter manifest.

☐ Every model/tokenizer/KB/index uses a local path; inference requires no external API.

☐ One command runs from the input directory to `output.zip`.

☐ Exactly 100 correctly named JSON files; no extras; UTF-8; deterministic ordering.

☐ Every entity satisfies `original_text[start:end] == text`.

☐ Every candidate exists in the frozen KB snapshot and belongs to the correct ontology for its type.

☐ Config, seed, Git commit, package versions, and hashes are logged.

☐ Golden tests and stress suites pass.

☐ VRAM/RAM/runtime profile is measured on an environment close to the organizer's machine.

☐ README includes troubleshooting and the expected directory tree.